

Lecture 9 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 9 extends the regression framework to categorical predictors, multiple factors, and interactions. The main idea is that group comparisons and ANOVA-style models can be written as regression models using indicator variables and Wilkinson notation. The lecture uses the Palmer penguins data to introduce categorical predictors and interactions, then uses birth-weight data to show how modeling choices differ for continuous and categorical predictors. It ends by connecting Bayesian model effect sizes to frequentist sums-of-squares measures: R^2 , partial R^2 , η^2 , and Cohen's f .

Lecture Sections

9.1 — Regression notation

This section introduces Wilkinson notation as a compact grammar for specifying regression models. The dependent variable appears on the left side of the formula, while independent variables and their combinations appear on the right. The penguin dataset provides the main running example, with body mass modeled using continuous predictors such as flipper length and categorical predictors such as species and sex.

New Definitions and Terms

- **Wilkinson notation** — a formula grammar for writing regression models, such as `body_mass_g ~ flipper_length_mm`

Formulas

$$y \sim x_1 + x_2$$

The left side gives the dependent variable, and the right side gives the predictors used to explain it.

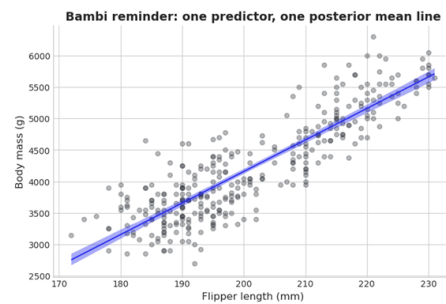
Figure

We can regress body mass on flipper length

- Variables we will use include:

- Body mass
- Flipper length
- Bill length
- Species
- Sex

```
bmb.Model(  
  "body_mass_g ~ flipper_length_mm",  
  data=penguins,  
)
```



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Figure 1: Slide 8 — We can regress body mass on flipper length

9.2 — Categorical predictors

This section shows that a categorical predictor can be written as a regression model by using indicator variables. In the penguin example, Adelie is the reference species, and the other species are represented by dummy variables. The intercept estimates the mean for the reference group, while the species coefficients estimate differences from that reference group.

New Definitions and Terms

- **Categorical predictor** — a predictor whose values are group labels rather than numerical measurements
- **Indicator variable** — a variable that equals 1 when an observation belongs to a category and 0 otherwise
- **Dummy coding** — representing categories with indicator variables
- **Reference coding** — dummy coding in which one group is represented by the intercept and all other groups are represented as differences from it
- **Effects coding** — coding in which dummy variables sum to 0 across groups
- **Cell means coding** — coding in which each group has its own parameter and there is no special reference intercept

Formulas

For species coded relative to Adelie penguins:

$$I_{1,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 1 & \text{if species}_i = \text{Chinstrap} \\ 0 & \text{if species}_i = \text{Gentoo} \end{cases} \quad I_{2,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 0 & \text{if species}_i = \text{Chinstrap} \\ 1 & \text{if species}_i = \text{Gentoo} \end{cases}$$

The model mean becomes:

$$\mu_i = \beta_0 + \beta_1 I_{1,i} + \beta_2 I_{2,i}$$

Equivalently:

$$\mu_i = \begin{cases} \beta_0 & \text{if species}_i = \text{Adelie} \\ \beta_0 + \beta_1 & \text{if species}_i = \text{Chinstrap} \\ \beta_0 + \beta_2 & \text{if species}_i = \text{Gentoo} \end{cases}$$

Figure

We can write categorical variables as a regression

- Use indicator variables
- Get a different combination of parameters for each group
- In Wilkinson notation
 - `body_mass ~ species`

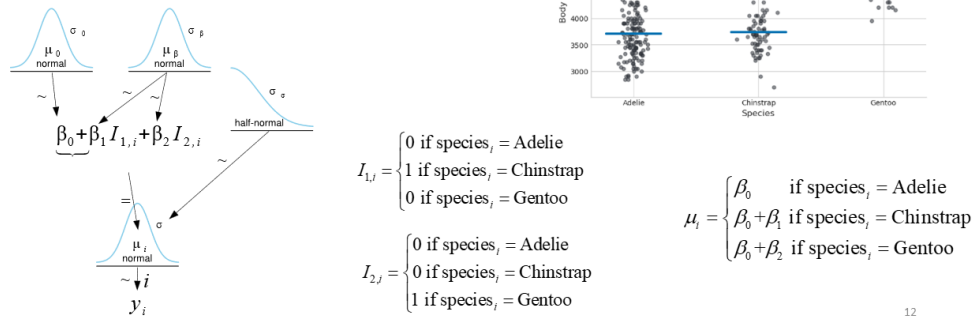


Figure 2: Slide 12 — We can write categorical variables as a regression

Coefficients are differences from Adelie penguins

- Gentoo penguins seem bigger

```
coef_summary = az.summary(idata_mass_species, var_names=["species"])
```

	mean	sd	hdi_3%	hdi_97%
species[Chinstrap]	25.296	67.623	-98.350	153.913
species[Gentoo]	1385.146	57.599	1279.879	1491.911

Group differences as coefficient differences

Chinstrap vs Adelie	species[Chinstrap]
Gentoo vs Adelie	species[Gentoo]
Chinstrap vs Gentoo	species[Gentoo] - species[Chinstrap]

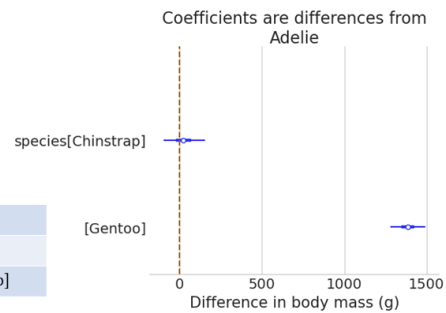


Figure 3: Slide 17 — Coefficients are differences from Adelie penguins

9.3 — Two categorical predictors

This section extends categorical regression to two predictors: species and sex. Bambi automatically creates dummy variables for both factors, so the model can be written with the same Wilkinson grammar as a continuous regression. The additive model assumes that the sex difference is the same for all species and that species differences are the same for both sexes.

New Definitions and Terms

- **Factor** — a categorical variable in a regression formula
- **Level** — one possible value of a factor
- **Additive model** — a model in which predictors contribute separate terms but do not change each other's effects
- **Main effect** — the contribution of one predictor in an additive model

Formulas

An additive categorical model with species and sex:

$$\mu_i = \beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} I_{\text{sex},i}$$

For sex coded as male relative to female:

$$I_{\text{sex},i} = \begin{cases} 1 & \text{if male} \\ 0 & \text{if female} \end{cases}$$

For a male and female penguin from the same species:

$$\mu_i - \mu_j = \beta_{\text{sex}}$$

This is the assumption that the sex difference is constant across species.

Figure

The additive model makes strong assumptions

- Sex differences are the same for all species

$$\mu_i = \beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} I_{\text{sex},i} \quad I_{\text{sex},i} = \begin{cases} 1 & \text{if male} \\ 0 & \text{if female} \end{cases}$$

Two dummy variables for species
One dummy variable for sex

For the difference between any male penguin i and female penguin j from the same species:

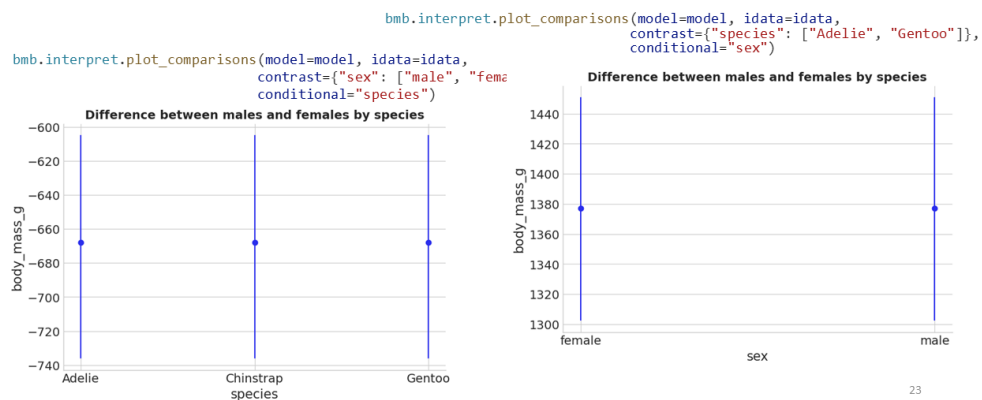
$$\begin{aligned} \mu_i - \mu_j &= \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} \right) - \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,j} \right) \\ &= \beta_{\text{sex}} \quad \text{because } I_{\text{species},k,i} = I_{\text{species},k,j} \text{ for all } k \end{aligned}$$

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Figure 4: Slide 21 — The additive model makes strong assumptions

No interaction between sex and species in the additive model

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Figure 5: Slide 23 — No interaction between sex and species in the additive model

9.4 — Interactions

This section introduces interactions as changing slopes. In an additive model with two continuous predictors, each predictor has a fixed slope. In an interaction model, the slope of one predictor depends on the value of the other predictor, which is represented by a product term in the regression mean.

New Definitions and Terms

- **Interaction** — a model term allowing the effect of one predictor to depend on another predictor
- **Interaction term** — a product term, such as x_1x_2 , added to the model
- **Correlated predictors** — predictors that carry overlapping information, making individual coefficients harder to interpret

Formulas

Additive model:

$$\mu_i = \beta_0 + \beta_1x_1 + \beta_2x_2$$

Interaction model:

$$\mu_i = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_{12}x_1x_2$$

Wilkinson notation:

$$y \sim x_1 + x_2 + x_1 : x_2$$

or equivalently:

$$y \sim x_1 * x_2$$

The `:` operator adds only the interaction term, while `*` adds the main effects and the interaction.

Figure

Interactions mean changing slopes

- For two continuous predictors
 - The additive model has a fixed slope for each one
- In an interaction model:
 - the slope of one variable depends on the value of the other
 - Written as an additive model with an interaction term

Additive model $\mu_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

Interaction model

$$\mu_i = \beta_0 + (\beta'_{1,0} + \beta'_{1,2} x_2) x_1 + (\beta'_{2,0} + \beta'_{2,1} x_1) x_2$$

$$= \beta_0 + \beta'_{1,0} x_1 + \beta'_{2,0} x_2 + (\beta'_{1,2} + \beta'_{2,1}) x_1 x_2$$

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

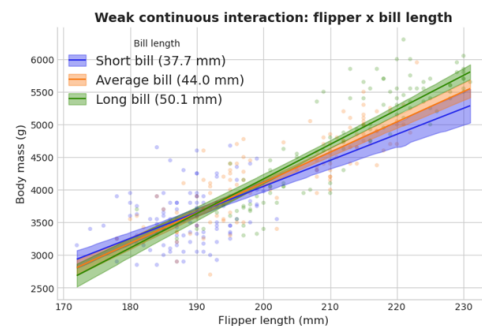
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Figure 6: Slide 25 — Interactions mean changing slopes

Fit a Bambi model to see the changing slope

- Specify the interaction model
 - And automatically get changing slope

```
bmb.Model(  
    "body_mass_g ~ flipper_length_mm * bill_length_mm",  
    data=penguins_cont,  
)  
  
bmb.interpret.plot_predictions(  
    model, idata,  
    conditional={  
        "flipper_length_mm": np.linspace(  
            penguins_cont["flipper_length_mm"].min(),  
            penguins_cont["flipper_length_mm"].max(),  
            80,  
        ),  
        "bill_length_mm": bill_level_values,  
    },  
)
```



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Figure 7: Slide 28 — Fit a Bambi model to see the changing slope

9.5 — Interacting categorical variables

This section applies interaction logic to categorical variables. Using `0 + species:sex` switches to a cell-means representation in which each species-sex combination receives its own mean. In the penguin data, the interaction is visible because the male-female body-mass difference is smaller for Chinstrap penguins and larger for Gentoo penguins.

The section also introduces the ToothGrowth example, where dose and supplement type interact. The posterior plot suggests that treating dose categorically may sometimes be more appropriate than forcing it to behave as a continuous linear predictor.

New Definitions and Terms

- **Categorical interaction** — an interaction between two categorical predictors
- **Cell mean** — the mean for one combination of factor levels

Formulas

Cell-means interaction model:

$$y \sim 0 + \text{species} : \text{sex}$$

This gives a separate parameter for each combination of species and sex.

Mixed categorical-continuous interaction:

$$\text{len} \sim \text{dose} * \text{supp}$$

This allows the effect of dose to differ between supplement types.

Figure

The interaction is visible by eye

- Sex difference
 - smaller in Chinstrap
 - And larger in Gentoo

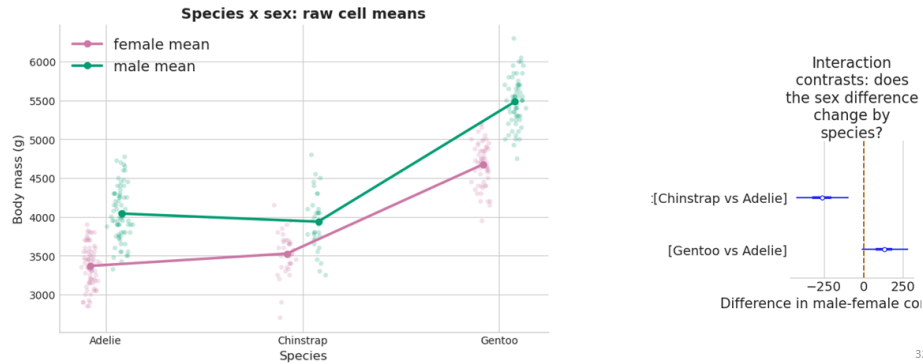


Figure 8: Slide 32 — The interaction is visible by eye

Although posterior plot suggests categorical might work better

$$\mu = \beta_0 + \beta_S S + \beta_D D + \beta_{SD} S D$$

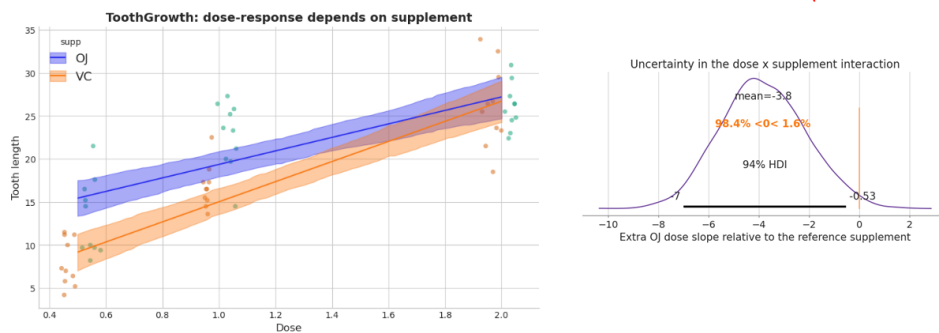


Figure 9: Slide 34 — Although posterior plot suggests categorical might work better

9.6 — Categorical vs continuous predictors

This section uses birth weight, gestation, and smoking to show that modeling choices are substantive, not just technical. A continuous gestation model assumes that a one-week change has the same effect at all gestational ages. A categorical gestation model relaxes that assumption, but it also creates unbalanced categories and wider posterior intervals.

The birth-weight example shows that an additive model suggests a possible smoking-related decrease, while the interaction model is less decisive. The categorical version clarifies that effects may vary with gestational term, but the HDIs also show that the available data are not enough to resolve every category-specific effect.

New Definitions and Terms

- **Continuous predictor** — a predictor treated as numerical, with changes interpreted per unit
- **Unbalanced categories** — categories with unequal sample sizes

Formulas

Additive birth-weight model:

$$\text{birth_weight} \sim \text{gestation} + \text{smoking}$$

Interaction birth-weight model:

$$\text{birth_weight} \sim \text{gestation} * \text{smoking}$$

Categorical gestation model:

$$\text{birth_weight} \sim 0 + \text{gestation_category} : \text{smoking}$$

Figure

Fit an additive model and see a possible difference

- Smoking introduces a decrease of 175 g (HDI: -260 to 50)

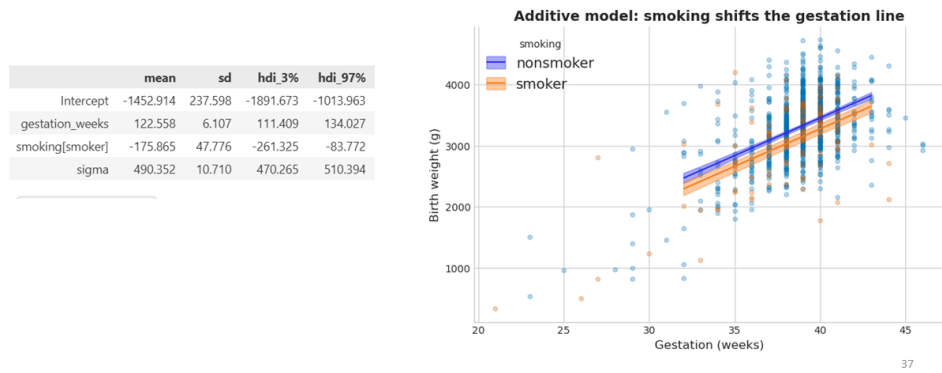


Figure 10: Slide 37 — Fit an additive model and see a possible difference

Effects may vary with gestational term

- HDIs suggest we don't have enough data
 - Should we repeat the study in a country with more smoking?
- Categorical coding clarifies the picture
 - In this case

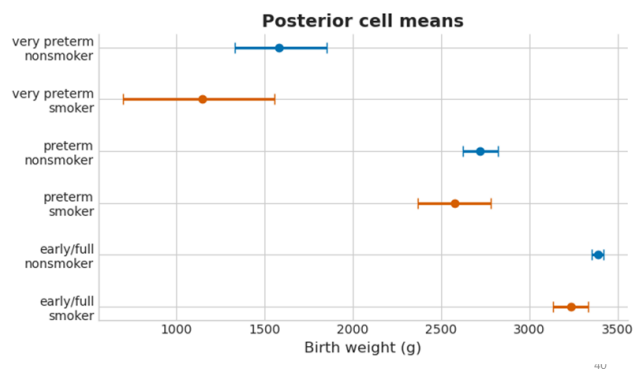


Figure 11: Slide 40 — Effects may vary with gestational term

9.7 — Model effect sizes

This section asks how much explanatory power a model or model term has. Bayesian R^2 is framed as variance in the model mean relative to the total modeled variance: explained variation plus residual noise. Partial R^2 , η^2 , and Cohen's f then measure the contribution of individual predictors or factors by comparing the full model to versions in which terms are left out.

The important distinction is the denominator. Partial R^2 asks how much of the remaining unexplained variance is explained by adding a term. η^2 compares the added variance to a total-variance scale, while Cohen's f expresses the added effect relative to the residual standard deviation.

New Definitions and Terms

- **Partial R^2** — variance explained by one term relative to what remains unexplained after considering other terms
- η^2 — variance explained by a factor relative to a total-variance scale
- **Cohen's f** — added explained variation expressed in units of residual variation

Formulas

Bayesian R^2 :

$$\text{var}(y) = \text{var}(\mu_i) + \sigma^2$$

$$R^2 = \frac{\text{var}(\mu_i)}{\text{var}(\mu_i) + \sigma^2}$$

Reduced means for a two-predictor interaction model:

$$\mu_i^{x_1} = \beta_0 + \beta_2 x_{2,i} + \beta_{12} x_{1,i} x_{2,i}$$

$$\mu_i^{x_2} = \beta_0 + \beta_1 x_{1,i} + \beta_{12} x_{1,i} x_{2,i}$$

$$\mu_i^{x_1 x_2} = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i}$$

Partial R^2 for a term:

$$R_{x_1}^2 = \frac{\text{var}(\mu_i^{x_1} - \mu_i)}{\text{var}(\mu_i^{x_1} - \mu_i) + \sigma^2}$$

Eta-squared for a term:

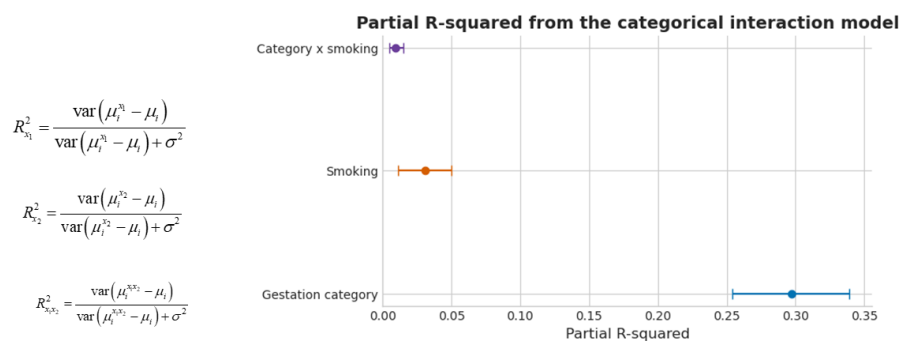
$$\eta_{x_1}^2 = \frac{\text{var}(\mu_i^{x_1} - \mu_i)}{\text{var}(\mu_i) + \sigma^2}$$

Cohen's f for a term:

$$f = \frac{\text{std}(\mu_i^{x_1} - \mu_i)}{\sigma}$$

Figure

Partial R^2 shows the contribution of each factor



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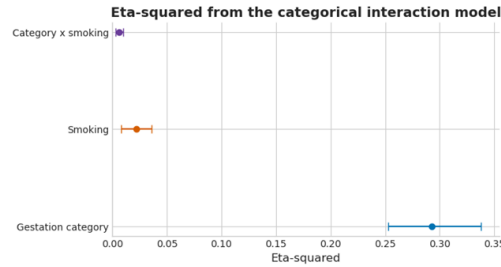
Figure 12: Slide 45 - Partial R-squared from the categorical interaction model

η^2 measures variance explained by a factor relative to its full model

- R^2 : Variance explained out of what was left
 - After considering all other factors
- η^2 : Variance explained relative to total variance

$$R^2_{\eta_1} = \frac{\text{var}(\mu_{i_1} - \mu_i)}{\text{var}(\mu_{i_1} - \mu_i) + \sigma^2}$$

$$\eta^2_{\eta_1} = \frac{\text{var}(\mu_{i_1} - \mu_i)}{\text{var}(\mu_i) + \sigma^2}$$



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Figure 13: Slide 46 - Eta-squared from the categorical interaction model

9.8 — Frequentist translation

This section translates the Bayesian effect-size discussion into frequentist sums of squares. Frequentist R^2 , partial R^2 , η^2 , and Cohen's f are calculated using fitted values, residual sums of squares, and total sums of squares. The key idea is still model comparison: the contribution of a term is measured by how much the predictions change when that term is removed.

Partial R^2 normalizes the added explained sum of squares by added explained variation plus residual variation. Partial η^2 normalizes added explained variation by the total variance for that factor plus noise. Cohen's f uses the same added explained variation but expresses it on a residual-standard-deviation scale.

New Definitions and Terms

- **Sum of squares** — a squared-deviation measure of variation
- **Added explained sum of squares** — the increase in explained variation when a term is included
- **Frequentist R^2** — explained sum of squares divided by total sum of squares

Formulas

Residual and regression sums of squares:

$$SS_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2$$

$$SS_{\text{reg}} = \sum_i (\hat{y}_i - \bar{y})^2$$

$$SS_{\text{tot}} = \sum_{i=1}^N (y_i - \bar{y})^2 = SS_{\text{reg}} + SS_{\text{res}}$$

Frequentist R^2 :

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}}$$

Added explained sum of squares for x_1 :

$$SS_{\text{reg},x_1} = \sum_i (\hat{y}_{x_1,i} - \hat{y}_i)^2$$

Here $\hat{y}_{x_1,i}$ is the predicted value from a model without x_1 .

Partial R^2 :

$$R_{x_1}^2 = \frac{SS_{x_1,\text{reg}}}{SS_{x_1,\text{reg}} + SS_{\text{res}}}$$

Partial η^2 :

$$\eta_{x_1}^2 = \frac{SS_{x_1,\text{reg}}}{SS_{\text{tot}}}$$

Cohen's f :

$$f = \frac{\sqrt{SS_{\text{reg},x_1}}}{\sqrt{SS_{\text{res}}}}$$

Figure

R^2, η^2 , and f are calculated in terms of sums of squares

$$\begin{aligned}
 SS_{\text{res}} &= \sum_i (y_i - \hat{y}_i)^2 \\
 SS_{\text{reg}} &= \sum_i (\hat{y}_i - \bar{y})^2 \\
 SS_{\text{tot}} &= \sum_{i=1}^N (y_i - \bar{y})^2
 \end{aligned}
 \qquad
 R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}}$$

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Figure 14: Slide 49 - R-squared, eta-squared, and f in terms of sums of squares

Partial R^2 is the added variance explained by a factor

- Normalized by the residual variance after adding all the other factors

$$\begin{aligned}
 SS_{\text{res}} &= \sum_i (y_i - \hat{y}_i)^2 \\
 SS_{\text{reg}} &= \sum_i (\hat{y}_i - \bar{y})^2 \\
 SS_{\text{tot}} &= \sum_{i=1}^N (y_i - \bar{y})^2 = SS_{\text{reg}} + SS_{\text{res}}
 \end{aligned}
 \qquad
 SS_{\text{reg}, x_1} = \sum_i (\hat{y}_{x_1, i} - \hat{y}_i)^2$$

$\hat{y}_{x_1, i}$ is the predicted value from a model without x_1

$$R_{x_1}^2 = \frac{SS_{x_1, \text{reg}}}{SS_{x_1, \text{reg}} + SS_{\text{res}}}$$

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Figure 15: Slide 50 - Normalized by the residual variance after adding all the other factors

Partial η^2 is the added variance explained by a factor

- Normalized by the total variance for that factor plus noise

$$\begin{aligned}
 SS_{\text{res}} &= \sum_i (y_i - \hat{y}_i)^2 \\
 SS_{\text{reg}} &= \sum_i (\hat{y}_i - \bar{y})^2 \\
 SS_{\text{tot}} &= \sum_{i=1}^N (y_i - \bar{y})^2 = SS_{\text{reg}} + SS_{\text{res}} \\
 \eta_{x_1}^2 &= \frac{SS_{x_1, \text{reg}}}{SS_{\text{tot}}}
 \end{aligned}$$

$\hat{y}_{x_1, i}$ is the predicted value from a model without x_1

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Figure 16: Slide 51 - Normalized by the total variance for that factor plus noise

Cohen's f is the added variance explained by a factor

- Normalized by the residual standard deviation

$$\begin{aligned}
 SS_{\text{res}} &= \sum_i (y_i - \hat{y}_i)^2 \\
 SS_{\text{reg}} &= \sum_i (\hat{y}_i - \bar{y})^2 \\
 SS_{\text{tot}} &= \sum_{i=1}^N (y_i - \bar{y})^2 = SS_{\text{reg}} + SS_{\text{res}} \\
 f &= \frac{\sqrt{SS_{\text{reg}, x_1}}}{\sqrt{SS_{\text{res}}}}
 \end{aligned}$$

$\hat{y}_{x_1, i}$ is the predicted value from a model without x_1

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Figure 17: Slide 52 - Normalized by the residual standard deviation